Model Question Paper: Science and General Studies

Time: 3 Hours Maximum Marks: 21

Instructions: All questions are compulsory. In Section A each question carries 1 mark.

Section A

1. Who wrote the national anthem of India?

(a) Rabindranath Tagore (b) Bankim Chandra Chatterjee

(c) Mahatma Gandhi (d) Sarojini Naidu

2. A train 375 m long passes a pole in 15 seconds. What is the speed of the train?

(a) 25 m/s (b) 15 m/s

(c) 20 m/s (d) 10 m/s

3. A car travels 200 km in 5 hours. What is its average speed?

(a) 64 km/h (b) 24 km/h

(c) 16 km/h (d) 40 km/h

4. Which planet is known as the Red Planet?

(a) Venus (b) Jupiter

(c) Saturn (d) Mars

5. A shopkeeper sells an article for Rs 1200 after giving a discount of 25%. What is its marked price?

(a) Rs 1600 (b) Rs 1280

(c) Rs 640 (d) Rs 1920

6. An object of mass 3600 kg has a volume of 4 cubic metres. What is its density?

(a) 1260 kg per cubic metre
(b) 720 kg per cubic metre
(c) 540 kg per cubic metre
(d) 900 kg per cubic metre

7. Which gas do plants absorb during photosynthesis?

(a) Nitrogen
(b) Hydrogen
(c) Carbon dioxide
(d) Oxygen

8. In which year did India become independent?

(a) 1948
(b) 1946
(c) 1945
(d) 1947

9. How many days are there in a leap year?

(a) 365
(b) 366
(c) 364
(d) 367

10. A train 150 m long passes a pole in 15 seconds. What is the speed of the train?

(a) 13 m/s
(b) 8 m/s
(c) 10 m/s
(d) 9 m/s

11. In which year was the Constitution of India adopted?

(a) 1950
(b) 1948
(c) 1947
(d) 1949

12. The average of 10 numbers is 22. If the number 38 is removed, what is the average of the rest?

(a) 20
(b) 28
(c) 16
(d) 12

13. The average of 5 numbers is 53. If the number 62 is removed, what is the average of the rest?

(a) 20 (b) 50

(c) 40 (d) 60

14. A shopkeeper sells an article for Rs 720 after giving a discount of 10%. What is its marked price?

(a) Rs 1120 (b) Rs 640

(c) Rs 800 (d) Rs 480

15. A shopkeeper sells an article for Rs 1440 after giving a discount of 10%. What is its marked price?

(a) Rs 1600 (b) Rs 2240

(c) Rs 1280 (d) Rs 960

Section B

16. Why does the sky appear blue? Explain briefly. [2 marks]

17. State Ohm's law and write its formula. [2 marks]

18. Explain the process of photosynthesis. [2 marks]